Route of entry – faecal oralroute (ingestion)

Incubation period – 7–14 days

The infection is acquired by ingestion of contaminated food and water

↓

The bacilli reaches the small intestine and attach to the epithelial cells of intestinal villi and penetrate to the lamina propria and submucosa

↓

These bacilli are phagocytosed by neutophils and macrophages. They resist intracellular killing and multiply within these cells.

↓

The pathogen enters the mesenteric lymph nodes, multiply there and enters blood stream through the thoracic duct

↓

A transient bacteraemia follows and internal organs like liver, gall bladder, spleen, bone marrow, lungs, lymph nodes and kidneys are infected

↓

The bacilli multiply abundantly in the gall bladder (bile juice) and are dischared continuously into the intestine involving peyer's patches and ileum.

↓

It becomes inflamed, necrosed and slough off, leaving behind typhoid ulcers. These ulcers may lead to two major complication, which are Intestinal perforation and haemorrhage.

Flowchart 7.6: Pathogenesis of *Salmonella typhi*

7.9.4 Clinical Manifestations

- The illness is usually gradual, with headache, malaise (feeling of discomfort), an2orexia (loss of appetite), coated tongue, abdominal discomfort with either constipation or diarrhea.
- Hepatosplenomegaly (enlargement of liver and spleen), step ladder pyrexia (continuous fever) and rose – spots (during 2nd or 3rd week).

7.9.5 Laboratory Diagnosis

Specimens: Blood, stool and urine are the clinical samples collected from typhoid patients. The selection of relevant specimen depends upon duration of illness, which is very important for diagnosis (Table 7.13 & Figure 7.15).

Table 7.13: Specimen collection for typhoid.

Duration of disease	Specimen examination	% Positivity
1st Week	Blood culture	90
2nd Week	Blood culture	75
	Faeces culture	50
	Widal test	Low titer
3rd Week	Widal test	80–100
	Blood culture	60
	Faeces culture	80

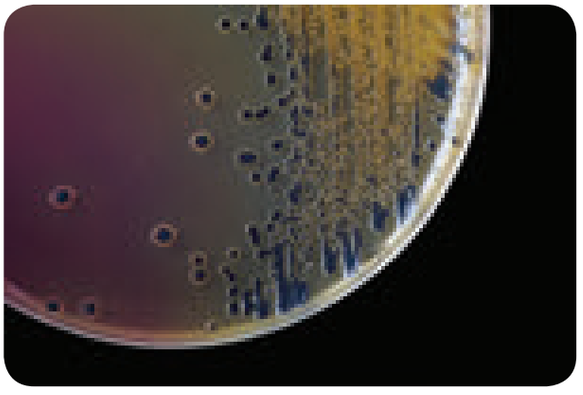


Figure 7.15: Colony morphology of *Salmonella typhi* on SS agar

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